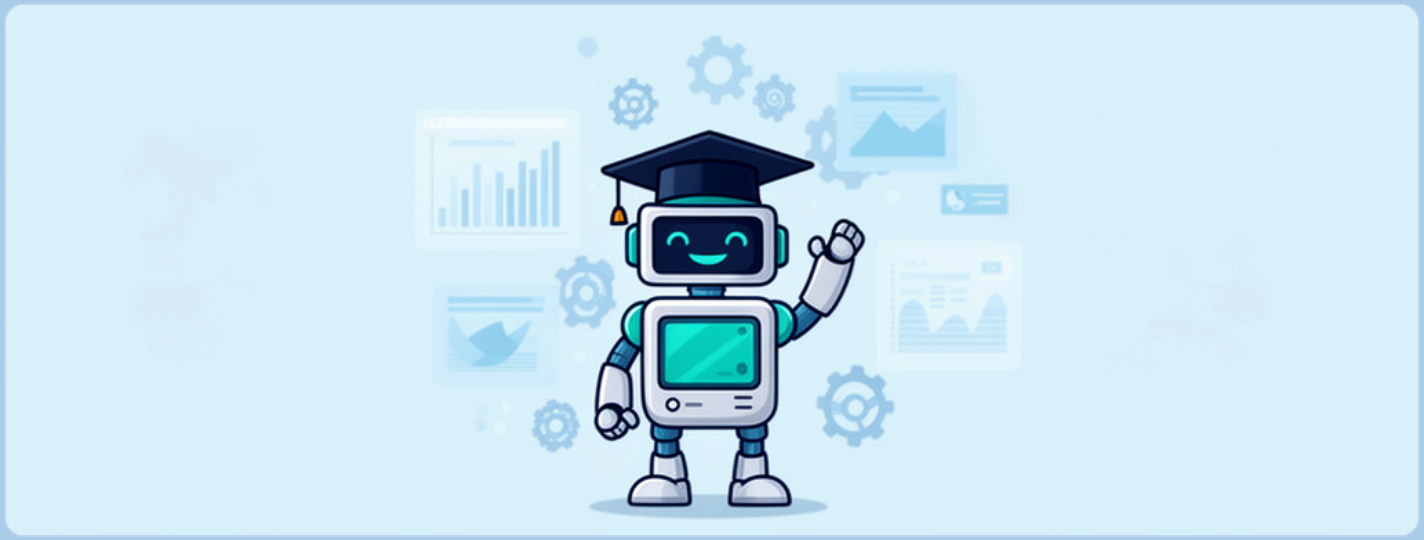
# ?What is Machine Learning

## Day One — You Build an AI

### Block 1 · Session 1

*.By the end of today, YOU will train a working ML model*



# The AI Special

# .One demo. One line. One question

*.Watch closely*

# The Animal Game

.The machine sees only this: **8 labeled animals**

Each animal = two numbers (weight, ear length) + one answer

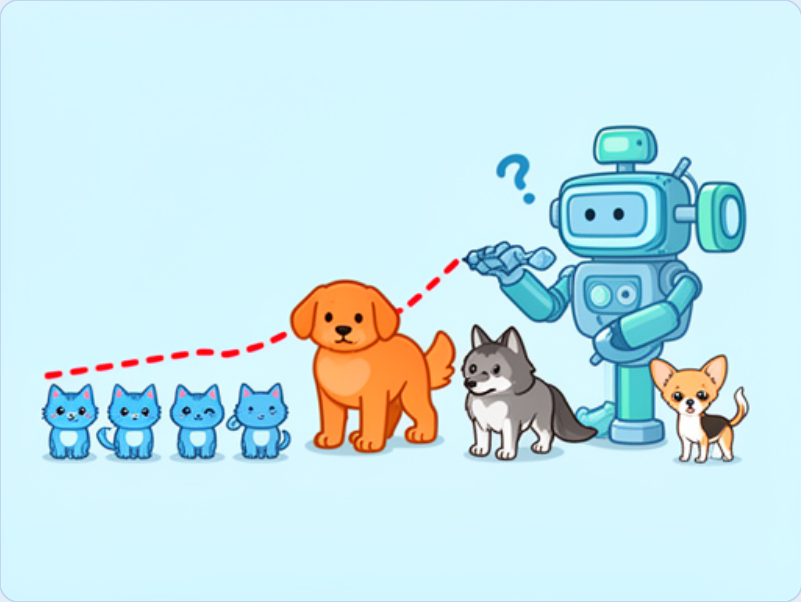
*.No rules were given to it. None*

# The Machine's Line

.From those 8 rows, the machine drew a line **all by itself**

.Behind the line: cats. In front: dogs

**"!?Who drew this line"**



# Now Let's Test It

:Animals the machine has **never seen**

!Wolf ← dog. Neighbor's cat ← cat. **Correct**

!?Puppy ← cat?! Elephant ← dog

*.One simple question fills the gaps in the line*

# The Big Question

## ?How do we make computers smart

# Way 1 — The Old Way: RULES

`data + RULES → answers`

.Humans write every rule by hand

"IF whiskers AND purrs THEN cat"

# The Problem with Rules

**.You try: write rules to recognize a cat**

!?Counterexamples welcome: hairless cats? cats in costumes? corgis

**.Humans can't write enough rules**



# Way 2 — Machine Learning

`data + answers → RULES`

**.We show examples. The machine finds the rules**

**.That red line at the start? This**

.Board box 2 under box 1: data + answers = RULES. Point silently — let it land

.CHECK (cold call): "What does the machine get and produce?" — examples with answers = rules

.Point back to these boxes at least 5 times today

<--

# AI vs ML

**AI** = the dream (machines acting smart)

**ML** = the main way we get there (learning from data)

# The Fruit Game

| Fruit ← | Texture | Size | Color |
|---|---|---|---|
| ? | smooth | medium | red |
| ? | smooth | long | yellow |
| ? | wrinkly | small | purple |

*.Cover the answer column. Guess the fruit from the clues alone*

# New Vocabulary

**Feature** = a clue *(a column)*

**Label** = the answer *(the column we predict)*

**Example** = one row = clues + answer

# ?Which Feature Matters

?Color? Size? Texture

**?Which clue is most useful for guessing the fruit**

*(.Vote. Debate. There is no wrong answer — yet)*

# Supervised Learning

.Learning from examples **that include the answers**

— Like studying with the answer key

.then facing a **new exam**

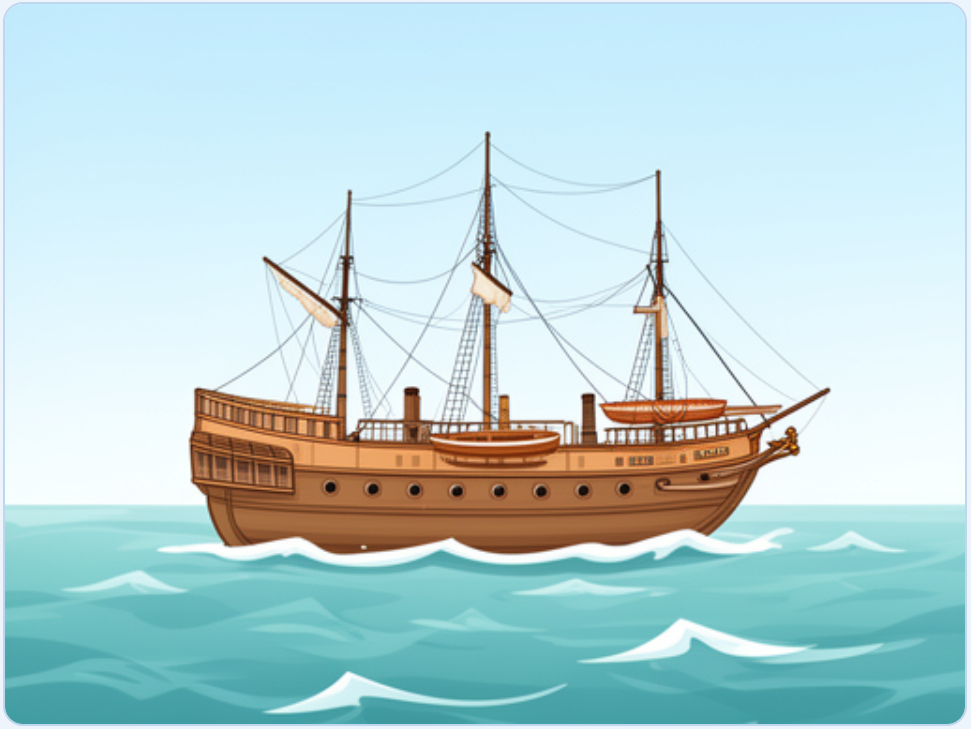
# ?"Why "Supervised

!No human is watching the model

.It means the **data contains the answers**

*( .History picked a weird word)*

# The Titanic

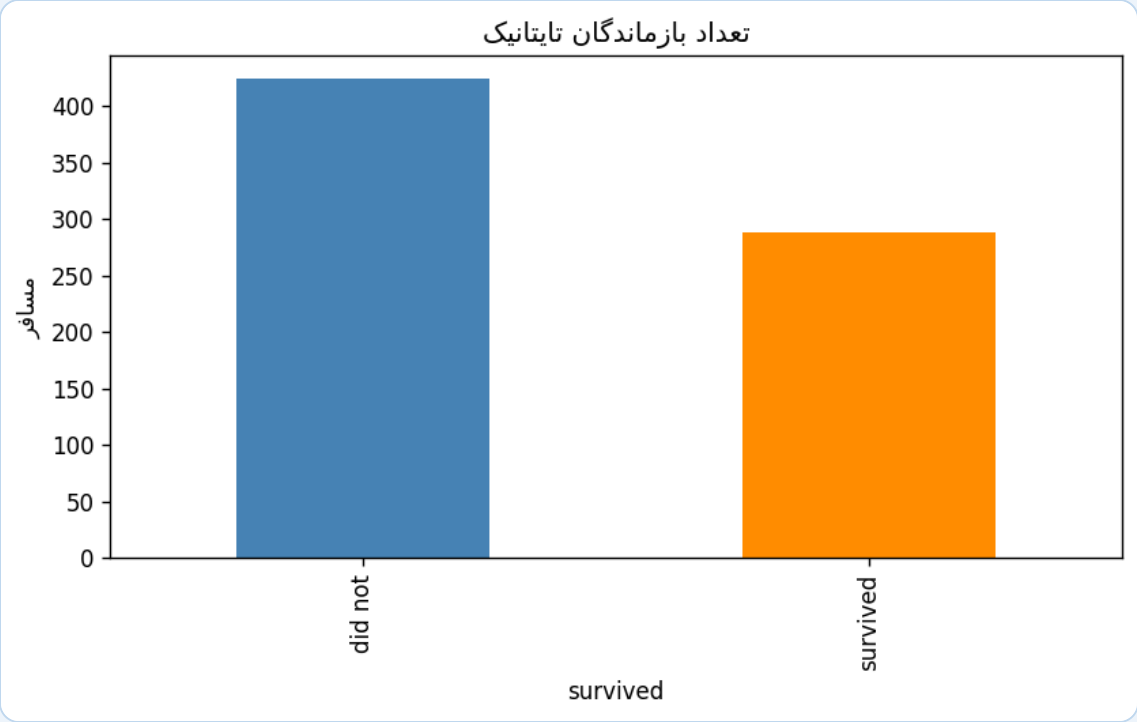


.passengers. 891 rows 891

.Every row is a life. Every column is a clue

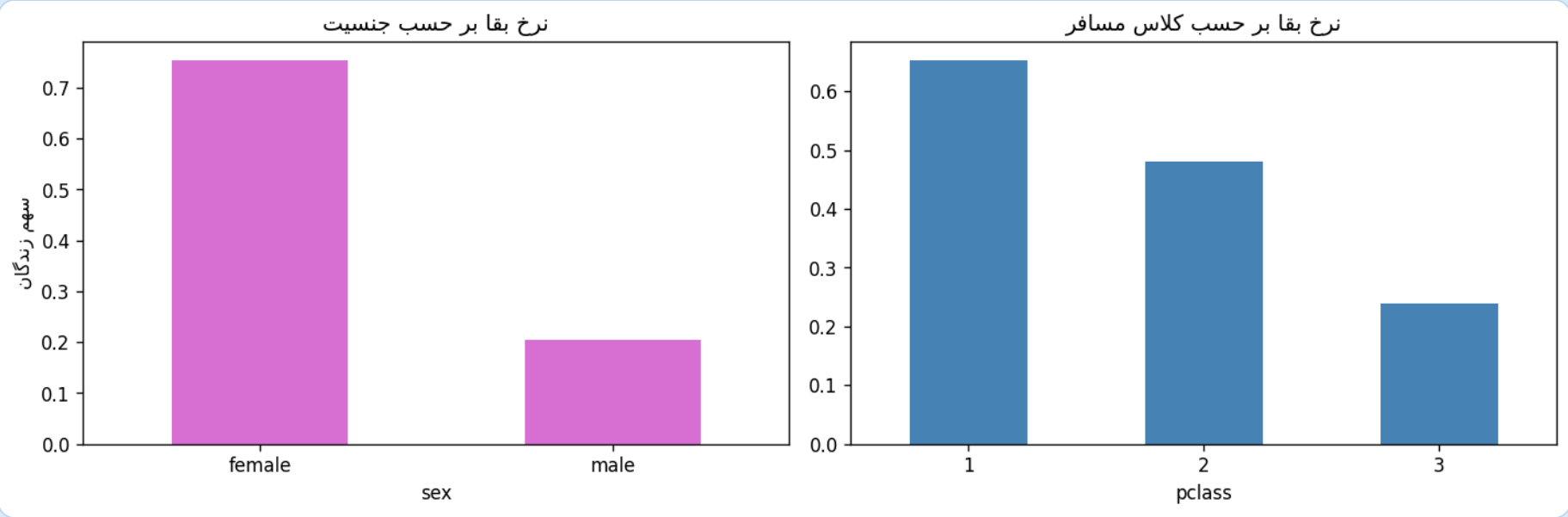
**?Would YOU have survived**

# Data as Pictures



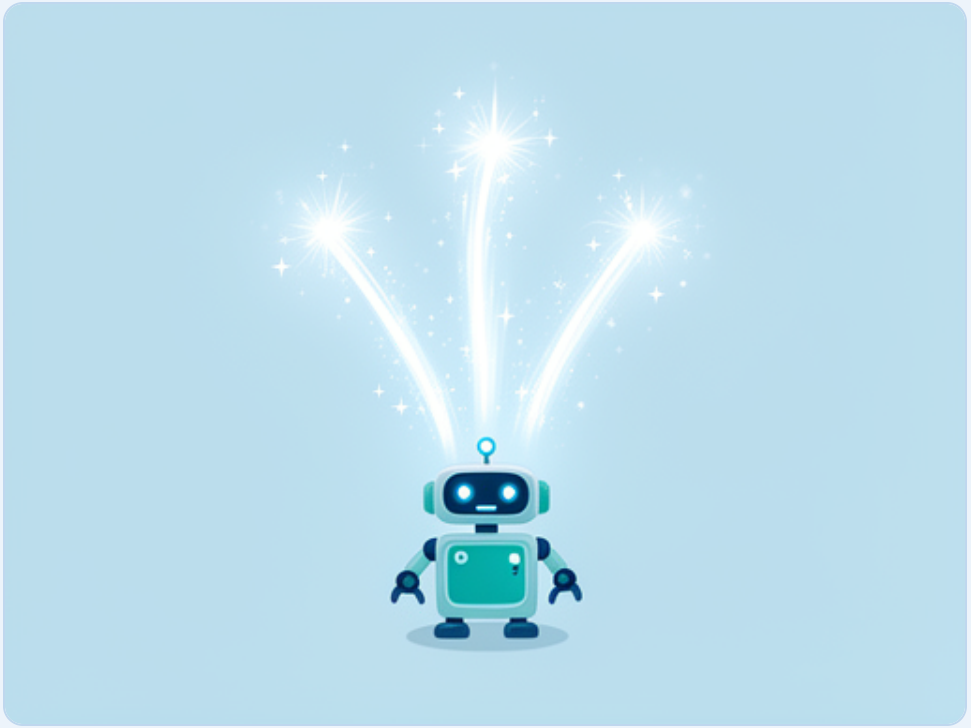

*.Always look at your data before doing anything with it*

# Survival by Sex & Class




.of women lived. ~19% of 3rd-class men did 74%~

# THE MOMENT



## Train an AI in 3 Lines

*.Type with me. Really — open the notebook and type*

# The 3 Magic Lines

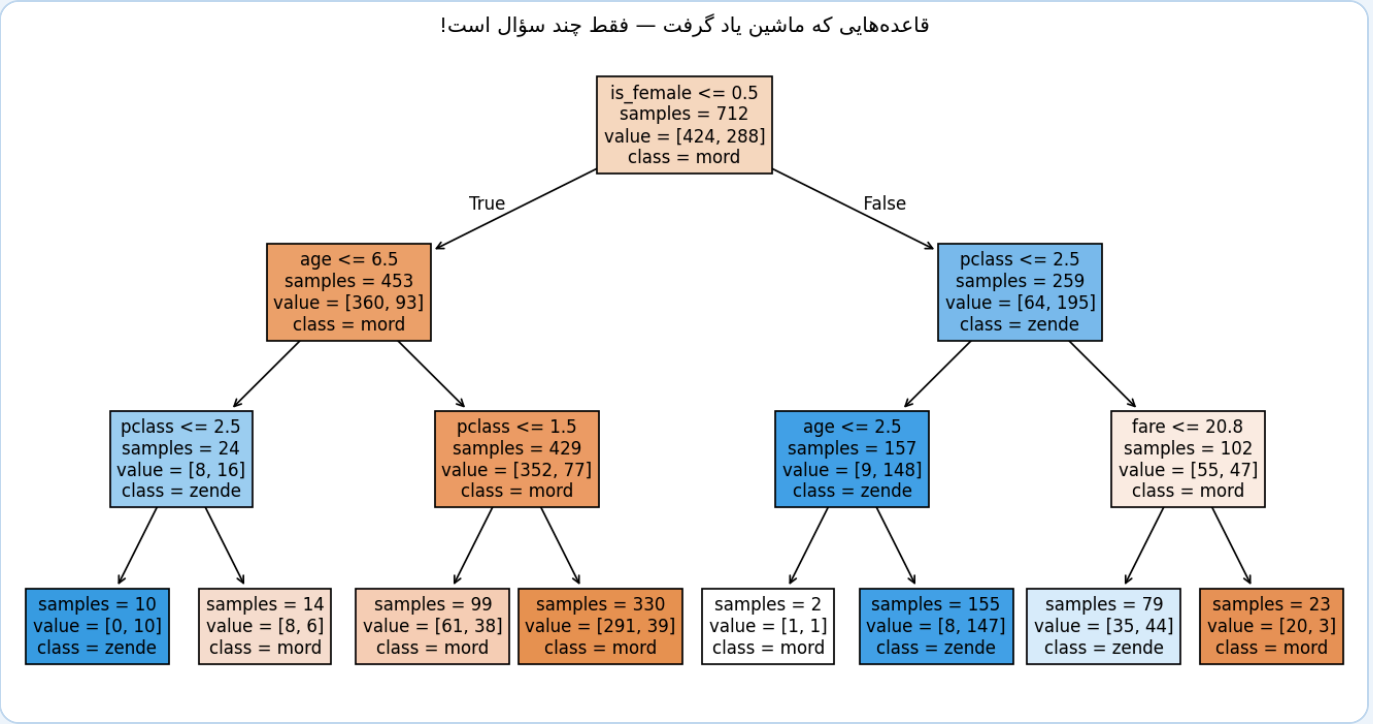
~/ml-course

```
model = DecisionTreeClassifier(max_depth=3) # 1. pick a student
model.fit(X, y) # 2. let it study
model.predict(new_passengers) # 3. ask it questions
```

**create → fit → predict**

*.This rhythm is ALL of ML. Not just today*

# What It Learned



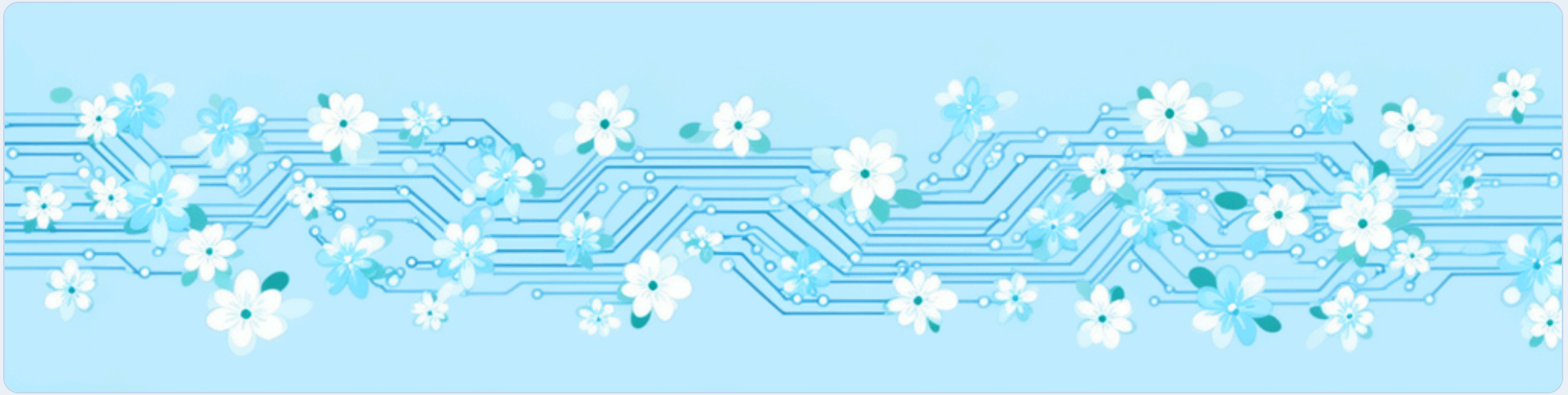

.It's just questions. You could draw this

# One Last Thing

.The model is **not a brain**

.It found question-patterns matching survival

**.That's already amazing enough**



# Today You Learned

- Rules → answers, or answers → **rules**
- **Features** (clues) & **labels** (answers)
- **Supervised** learning = examples with answers
- magic lines: **create** → **fit** → **predict** 3

# Your Ship Tonight

Your first ML notebook → **your GitHub repo**

**Would I have survived the Titanic?"** — an app you built**"**

# At-Home Problems (4)

1. **B1** The Feature Detective
2. **B2** Build Your Own Dataset
3. **B3** The 3 Magic Lines — train YOUR classifier
4. **B4** The Survival Predictor — a boy from 1912 awaits your verdict

**Due: before our problem-solving session.**

# Next Time

## .We solve YOUR problems — live

**.Bring your attempts. Bring your bugs**

*.They are the fuel of the next session*

